Aftab Akram

Antibes, France • +33(0)7 58 71 19 60 • aftab.akram@eurecom.fr • aftab201.github.io • Aftab201 • Google Scholar • DBLP • LinkedIn

EDUCATION

- **PhD in Cryptography and Security**, *Eurecom / University of Sorbonne* — **Biot, France** *2023-2026**

  Designed cryptographic protocols for privacy-preserving federated learning using Homomorphic Encryption, Secure Multiparty Computation, and threshold secret sharing, enabling secure aggregation with formal privacy guarantees in heterogeneous settings [AHMC25]. Extends this to fairness-aware secure aggregation, incorporating differential privacy during training to achieve formal ε-DP guarantees while preserving equitable model utility across heterogeneous clients [TAIM26]. Developed privacy-preserving federated unlearning protocols resistant to inference attacks [AFCM26], and Byzantine-robust aggregation using TEE-backed privacy-preserving smart contracts for tamper-evident computation resilient to malicious participants [ACMD+25].

- **M.Sc. in Information Security**, *National University of Science And Technology (NUST)* — **Islamabad, Pakistan** *2019-2021*

  The program covered a broad range of subjects. I chose Cryptography as my major, with topics including symmetric and asymmetric cryptography, number theory, algebra, finite-field algorithmics, complexity theory, post-quantum cryptography, zero-knowledge proofs, and more.

- **Bachelor in Electrical Engineering**, *Government College University* — **Lahore, Pakistan** *2016-2019*

  Specialized in Electronics and Telecommunications and graduated with strong academic performance. As part of the final-year project, designed and developed an Arduino-based wireless rescue robot with live video transmission and ultrasonic obstacle detection.

WORK EXPERIENCE

- **Teaching and Research Assistant**, *Eurecom* — **Biot, France** *2023-2026*

  Mentored and supervised Master's students on semester and final-year thesis projects in cybersecurity, privacy, and machine learning. Guided students in project design, implementation, and evaluation, monitored research progress, assessed technical reports and presentations, and served on examination and project evaluation committees.

- **Research Intern**, *National University of Sciences and Technology (NUST)* — **Islamabad, Pakistan** *2022-2023*

  Researched post-quantum cryptographic schemes, particularly CKKS and BFV homomorphic encryption, for privacy-preserving machine learning. Developed and optimized secure neural network inference frameworks over encrypted data, enabling efficient and confidential AI deployment in privacy-sensitive environments.

- **Management Associate**, *Pakistan Telecommunication Company Limited (PTCL)* — **Lahore, Pakistan** *2016–2017*

  Supported telecommunications operations and service delivery activities through performance monitoring, data analysis, and cross-functional coordination. Assisted in network and operational reporting, process optimization, and project management tasks while collaborating with technical and business teams to improve service quality and operational efficiency.

SKILLS

- **Cryptography**

  Fully Homomorphic Encryption (CKKS, BFV, post-quantum variants), Secret Sharing & Threshold Cryptography (Shamir), Secure Multi-Party Computation, Differential Privacy, Zero-Knowledge Proofs (foundational, growing focus)

- **Privacy & Trustworthy ML**

  Secure Aggregation, Byzantine-Robust Aggregation, Federated Learning, Federated Unlearning, Privacy-Utility Tradeoff Evaluation, Adversarial Robustness

- **ML, Blockchain & Computing**

  PyTorch, Flower, PySyft, Secret Network, Privacy-Preserving Smart Contracts, TEE-based Confidential Computing

- **Programming & Tools**

  Python (proficient), Rust (basic, actively developing), Git, Linux, LaTeX

PUBLICATIONS

[AFCM26] **Privacy-Preserving Federated Unlearning**, *Akram, Aftab; Alborch, Ferran; Gritti, Clémentine; Önen, Melek . Manuscript under review, AsiaCCS 2027*

[TAIM26] **Privacy-Preserving and Fairness-Aware Secure Aggregation in Federated Learning**, *El Bacha, Tonia; Akram, Aftab; Harrouche, Ibtissam; Önen, Melek*. Accepted at *Privacy in Statistical Databases (PSD 2026) Conference*, 2026.

[AHMC25] **SAAFL: Secure Aggregation for Label-Aware Federated Learning**, *Akram, Aftab; Pham, Harry N. H.; Önen, Melek; Gritti, Clémentine*. In *IFIP International Conference on ICT Systems Security and Privacy Protection*, Springer, 2025.

[ACMD+25] **Robust Blockchain-Based Federated Learning**, *Akram, Aftab; Gritti, Clémentine; Halip, Mohd Hazali Mohamed; Kamarudin, Nur Diyana; Mansor, Marini; Rahayu, Syarifah Bahiyah; Önen, Melek*. In *ICISSP 2025 — 11th International Conference on Information Systems Security and Privacy*, 2025.

[AFSA+24] **Privacy-Preserving Inference for Deep Neural Networks: Optimizing Homomorphic Encryption for Efficient and Secure Classification**, *Akram, Aftab; Khan, Fawad; Tahir, Shahzaib; Iqbal, Asif; Shah, Syed Aziz; Baz, Abdullah*. In *IEEE Access*, vol. 12, pp. 15684–15695, 2024.

OTHER CONTRIBUTIONS

[APCG+25] **Fundamentals of Privacy-Preserving and Secure Machine Learning** *(Book Chapter)*, *Akram, Aftab; Zimmer, Pascal; Gritti, Clémentine; Karame, Ghassan; Önen, Melek*. In *Trustworthy AI in Medical Imaging*, pp. 385–409. Academic Press, 2025.

OPEN SOURCE CONTRIBUTIONS

- **Robust Blockchain-Based Federated Learning**,

  2024

  Designed and implemented a blockchain-based secure aggregation framework for federated learning with a **Python**-based client/interface layer and **Rust**-based Secret Network smart contracts running in Trusted Execution Environments (TEEs). GitHub

- **SAAFL : Secure Aggregation for Label-Aware Federated Learning**,

  2024

  Designed and developed SAAFL, a privacy-preserving secure aggregation protocol for non-iid data based federated learning. GitHub

- **SaFer: Privacy-Preserving Fair Federated Learning**,

  2025

  Designed and implemented a secure aggregation framework enabling privacy-preserving and fairness-aware federated learning.GitHub

CERTIFICATIONS

Neural Networks and Deep Learning, Certificate
DeepLearning.AI

PRESENTATIONS

Conferences, Workshops, Forums

Sep. 2026* — **Privacy-Preserving and Fairness-Aware Secure Aggregation in Federated Learning**
Conference PSD'25, Cadiz, Spain

May 2025 — **Privacy-Preserving Federated Learning**
Scientific council annual workshop, Eurecom, France

Oct. 2025 — **Designing a Cryptographic Suites Scanner for Network Security Protocols**
REDOCS'23, Marseille, France

May 2025 — **SAAFL: Secure Aggregation for Label-Aware Federated Learning**
Conference IFIP SEC'25, Maribor, Slovenia

Feb. 2025 — **Robust Blockchain-Based Federated Learning**
Conference ICISSP'25, Porto, Portugal

Conference Organizer

June 2024 — **Privacy in Statistical Databases (PSD 2024) Conference**
PSD24, Antibes Juan-les-Pins, France

STUDENT SUPERVISION

- **Blockchain-based Federated learning**, *M. Aziz Sghaier, Thomas Du*
  Master in Computer Science at EURECOM — *2023-2024*
- **Blockchain-based Federated learning**, *Pierre DONTENVILLE, Guillaume BLANC*
  Master in Computer Science at EURECOM — *2024*
- **Data Heterogeneity based Federated Learning**, *Malek Sfaxi, Harry Pham*
  Master in Computer Science at EURECOM — *2024-2025*
- **Fairness-aware Federated Learning**, *Tonia El Bacha*
  Master in Computer Science at EURECOM — *2025-2026*

LANGUAGES

- **Urdu** : Native Speaker
- **English** : Fluent
- **French** : Basic
- **Arabic** : Basic

REFEREES

- **Melek Önen**, *Professor classe 2, Eurecom*
  Email: melek.onen@eurecom.fr. PhD supervisor
- **Clémentine Gritti** , *Assistant Professor, INSA Lyon*
  Email: clementine.gritti@insa-lyon.fr. PhD co-supervisor